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Phishing Website Detection using ML

Introduction

With the rapid growth of the internet, phishing attacks have become one of the most common cybersecurity threats. Phishing websites are designed to look like legitimate websites to trick users into revealing sensitive information such as passwords, credit card details, and personal data.

This project focuses on building a machine learning model that can detect whether a given website URL is **legitimate or phishing** based on various URL-based features. The system helps users stay safe by providing early warnings about suspicious websites.

Objective

- To develop a machine learning model for phishing website detection
- To extract and analyze URL-based features
- To classify websites as **safe or phishing**
- To create a simple interface for user input and prediction
- To apply AI/ML concepts learned in the course

Dataset Description

The dataset used contains information about various websites and their characteristics. It includes features extracted from URLs such as:

- Length of URL
- Presence of special characters (@, -, etc.)
- Use of HTTPS
- Number of subdomains
- Presence of suspicious keywords
- Domain-related features

Each entry is labeled as:

- **1 → Legitimate Website**
- **0 → Phishing Website**

Tools and Technologies Used

- **Python**
- **Pandas** – Data handling
- **NumPy** – Numerical operations
- **Scikit-learn** – Machine learning model
- **Pickle – Model saving/loading**
- **Streamlit / CLI (if used)** – User interface

Methodology

Step 1: Data Collection

- Dataset collected from online sources (e.g., Kaggle or phishing datasets)

Step 2: Feature Extraction

- Extracted important features from URLs
- Converted URLs into numerical format for model input

Step 3: Data Preprocessing

- Cleaned dataset
- Removed unnecessary data
- Ensured proper formatting

Step 4: Model Training

- Split dataset into training and testing sets
- Used **Logistic Regression / Random Forest**
- Trained model to classify phishing vs legitimate

Step 5: Model Evaluation

- Tested model accuracy
- Achieved accuracy of around **85–95%** (adjust if needed)

Step 6: Prediction System

• Created scripts (predict.py and run.py)
• run.py acts as the entry point for execution
• User runs the program using a single command
• User enters a URL
• Model predicts whether it is safe or phishing

System Architecture

User Input (URL)
↓
Feature Extraction
↓
Machine Learning Model
↓
Prediction Output (Phishing / Legitimate)

This system is based on a machine learning model and is still under development. Predictions may not always be 100% accurate. Users are advised to verify results before making decisions.

How to Run the Project (Execution Steps)

To run the Phishing Website Detector, follow these steps:

Step 1: Clone the Repository

```
git clone https://github.com/jiya25bai10733-gif/phishing-website-detector
cd phishing-website-detector
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Step 3: Run the Program

```
python run.py
```

Step 4: Enter URL

- Enter a website URL when prompted
- Example:

```
http://example.com
```

Step 5: View Result

- Output will be:
  - Phishing Website
  - or Legitimate Website

Note:
The project runs using a command-line interface (CLI) and does not depend on any graphical interface.

Project Requirements

- Python 3.x
- Required libraries:
  - pandas
  - numpy
  - scikit-learn
  - joblib

Results

- The model successfully detects phishing websites
- Provides instant prediction based on URL
- Accuracy achieved: **~85–95%**
- Simple and easy-to-use system

Challenges Faced

- Extracting meaningful features from URLs
- Handling missing or inconsistent data
- Model warnings (feature mismatch issues)
- Debugging file path and execution errors
- Understanding ML model behaviour

Future Scope

- Improve accuracy using advanced models (XGBoost, Deep Learning)
- Add **real-time browser extension**
- Integrate blacklist APIs
- Build a full web application
- Add visual dashboard for analysis

Conclusion

This project demonstrates how machine learning can be effectively used in cybersecurity to detect phishing websites. By analysing URL patterns and features, the system can identify malicious websites and help users avoid online scams.

It highlights the importance of feature engineering, model training, and real-world application of AI/ML concepts.

References

- Kaggle Phishing Dataset
- Scikit-learn Documentation
- Python Documentation
- Research papers on phishing detection

Note: This system is based on a machine learning model and is still under training/improvement. Predictions may not always be 100% accurate. Users are advised to verify results before making decisions.